Master Dataset Report

Complete End-to-End Methodology — Acquisition, Validation, Fusion

METHODOLOGICAL REPORT

Master's Thesis

Winter Wheat Harvest Window Prediction via Machine Learning

Centre-Val de Loire · France

Indrashil University — 2025

Supervisors: Dr. Vishvjit Thakar, Dr. Doreid Ammar

Version 1.0 — Methodological reference document

Contents

Intended audience

This methodological report is intended for:

- 👨‍🎓 **The thesis jury** and academic supervisors
- 👨‍💻 **Future data scientists** who will continue the project
- 👨‍🔬 **Researchers** who want to replicate or extend the methodology
- 📊 **Kaggle evaluators** assessing the scientific quality of the dataset
- 🎓 **Students** discovering agricultural remote sensing

> **Associated documents:** this report is designed to be read after the **Agronomic Introduction** (concepts) and in parallel with the **Master Data Dictionary** (technical schema). Together, these three documents form the complete project documentation.

PART I — Context and motivation

1. Scientific and applied problem

1.1 Research question

Question: Is it possible to **predict the optimal harvest date of winter wheat** at the parcel scale, several days in advance, by combining multi-source satellite data (optical + radar), climate data, and soil properties, using machine learning models?

1.2 The agronomic stake

Wheat harvesting is a **critical** step in the agricultural cycle. The exact harvest moment influences:

- **Grain quality:** moisture, protein content, baking strength
- **Final yield:** losses in case of late harvest (shedding, lodging, sprouting on the stalk)
- **Operating costs:** drying (if grain too wet), combine harvester planning
- **Logistics:** silos, transport, downstream processing

Harvesting **5 days too early** can mean a 10-15% quality loss and a drying cost of 20-30 €/ton. **5 days too late**, in case of storm or heatwave, can destroy 20-40% of yield.

1.3 The scientific stake

Beyond the direct agricultural stake, the project addresses several open research questions:

1. **Which satellite signals are most predictive** of maturation? Optical (NDVI)? Radar (σ^0)? The combination?
2. **What is the minimum temporal resolution** required for reliable prediction? 1 week? 2 weeks?
3. **Are soil and climate sufficient without satellite,** or do vegetation indices provide critical information?
4. **Which ML architecture is best suited?** Decision trees? Recurrent networks? Transformers?
5. **Can the model be transferred** to other regions or other crops?

1.4 The societal stake

Beyond direct agricultural stakes, this type of project contributes to:

- **Climate change adaptation:** tracking modifications of the agricultural calendar
- **Food sovereignty:** national harvest forecasting
- **Commodity markets:** transparency of forecasts
- **Open data in research:** public availability of data and methods

2. State of the art in agricultural remote sensing

2.1 Historical evolution

Period	Dominant approach	Limitations
Before 2000	Field observations + low-resolution satellites (NOAA AVHRR ~1km)	Insufficient resolution for parcels
2000-2015	Landsat (30m), MODIS (250m) — agronomic models	Slow revisit, cloud dependency
2015-2020	Sentinel-2 (10m, 5 days) — parcel-scale democratization	Still cloud-sensitive
2020+	**Multi-source (S2 + S1) + advanced ML/DL**	Still few public studies in France

2.2 Comparable works

Several recent studies have explored prediction of phenological stages or harvest:

Study	Area	Approach	Limitation
Vannoppen et al. (2022)	Belgium (Flanders)	S2 + LSTM for phenology	No SAR, regional scale
Lobert et al. (2023)	Germany	S2 + S1 for crop recognition	Classification, not date prediction
Selena Jasmine (2025)	India (rice)	S2 + EOS-06 SCAT-3 for yield	District scale, not parcel-level
Belda et al. (2020)	Spain	S2 for wheat phenology monitoring	No harvest date prediction

2.3 Gaps identified in the literature

To our knowledge, no public study simultaneously combines:

- **Parcel-level** scale (not regional or department)
- **Sentinel-2 + Sentinel-1** combination at parcel scale
- Integration of **soil + weather + satellites** in multi-source
- Application to **winter wheat in metropolitan France**
- **Public** availability of the complete dataset
- Prediction target = **harvest date** (not yield)

This combination forms the originality of the present master's thesis project.

3. Originality of our approach

3.1 Three methodological innovations

Innovation 1 — Multi-source at parcel scale

Almost all existing studies operate at regional or department scale. Our approach descends to the **individual agricultural parcel** (1500 parcels), which:

- Allows capturing **intra-regional variability** (different soils, micro-climates)
- Produces a richer training signal (1500 observations vs ~6 regional)
- Opens the way to parcel-level operational applications

Innovation 2 — S2 + S1 + soil + weather fusion

The project integrates 4 very different information modalities:

- **Optical (S2):** photosynthesis, photosynthetic vigor
- **Radar (S1):** canopy structure, biomass, cloud-independent
- **Soil (SoilGrids):** water retention capacity, chemical fertility
- **Weather (NASA POWER):** temperatures, rainfall, radiation, ET0

This quadruple integration is uncommon in the public literature.

Innovation 3 — Open data provision

The 6 raw datasets are deposited publicly on Kaggle under CC BY 4.0 license. This enables:

- **Total replicability** of the experiment
- **Reuse** by the scientific community
- **Benchmarking** of future methods
- **Academic value** (potential data paper)

3.2 Scientific positioning

Our contribution: to our knowledge, the project produces the **first multi-source public dataset at parcel scale for winter wheat harvest date prediction in France**. This original character opens publication prospects (data paper) in parallel with ML modeling results.

PART II — Global methodology

4. End-to-end pipeline

4.1 Pipeline overview

The project breaks down into **5 major phases**:

PHASE 1 — Sample selection (RPG)
300 random parcels × 5 years = 1500 parcels (seed=42, >1 ha)

▼

PHASE 2 — Multi-source acquisition
🎯 Céré'Obs · 📍 RPG · 🌦️ NASA POWER · 🪨 SoilGrids · 🛰️ Sentinel-2/-1
Public APIs · Google Earth Engine · Python tools

▼

PHASE 3 — Validation and documentation
EDA · plausibility checks · 19 documentation PDFs · Kaggle deposits

▼

PHASE 4 — Master Dataset assembly (upcoming)
Derived features · fusion on (parcelle_uid, year) · ~1500 × 200

▼

PHASE 5 — Machine Learning modeling (upcoming)
RF, XGBoost, LSTM, Transformer, multimodal · LOYO cross-validation

Figure 4.1 — End-to-end project pipeline: 5 distinct phases. The present report version documents phases 1 to 3 (completed). Phases 4 (assembly) and 5 (ML) will be documented in a future version.

4.2 Current project status

SOURCES ACQUIRED	KAGGLE DATASETS	PDF DOCUMENTS	TOTAL DATA VOLUME
6 / 6	**6 / 6**	**19**	**~620 MB**

5. Major methodological choices

Throughout the project design, several strategic decisions were made. This section documents them with their rationale.

5.1 Decision: tabular approach (not images)

Decision: aggregate all satellite data into **tabular features** (instead of using raw images).

Rationale:

- CNN/ViT on satellite images require expensive GPUs and massive data volumes
- The tabular approach allows using **all ML models** (RF, XGBoost, LSTM, Transformer)
- Vegetation indices (NDVI, EVI, NDWI) already condense the relevant physical information
- Interpretability (SHAP, LIME) is more direct on tabular features
- Our sample size (1500) is better suited to tabular models

5.2 Decision: 5 independent years (no inter-annual time series)

Decision: re-sample 300 new parcels each year (instead of tracking the same 300 parcels over 5 years).

Rationale:

- Crop rotation in CVL means only **17.1%** of parcels carry wheat in two consecutive years
- Following the same parcels would have drastically reduced the sample size
- For harvest date prediction, inter-annual continuity is not essential (each season is a complete cycle)
- This approach captures more agronomic diversity (different soils, farmers, practices)

5.3 Decision: Sentinel-1 + Sentinel-2 combined (not just one)

Decision: include Sentinel-1 (radar) in addition to Sentinel-2 (optical).

Rationale:

- Sentinel-2 loses ~26% coverage in winter due to clouds
- Sentinel-1 provides ~8 scenes per parcel per month **year-round**
- The VV/VH ratio is a **direct harvest marker** (sharp rise post-harvest)
- Combination reaches ~18 scenes/parcel/month, ideal for sequential models (LSTM, Transformer)

5.4 Decision: 3 temporal granularities (15j, 7j, raw)

Decision: export Sentinel-2 at 3 different temporal granularities (15-day composites, 7-day, and raw per scene).

Rationale:

- Different ML architectures require different temporal resolutions
- **15 days:** classical ML (RF, XGBoost) — dense data, few NaNs
- **7 days:** sequential models (LSTM, GRU) — finer resolution
- **Raw:** Transformer + total flexibility in Python
- Avoids relaunching Google Earth Engine for each tested architecture

5.5 Decision: WCS extraction (not REST API) for SoilGrids

Decision: download complete SoilGrids rasters via WCS instead of the point-by-point REST API.

Rationale:

- REST API: ~5 hours for 1500 parcels
- WCS rasters: ~3.3 minutes (**91× faster** gain)
- Also enables global raster visualization and validation
- Preserves flexibility to extract other parcels later without reloading

5.6 Decision: no private farmer data

Decision: use only **public and externally observable** data (no cultivated variety, sowing date, fertilization, etc.).

Rationale:

- Reproducibility: the method can be applied to any parcel without consent
- Ethics: no access to sensitive farmer data
- Scalability: applicable to millions of parcels without field collection

- This is also a methodological challenge: can we predict harvest without knowing the variety?

6. Tools and platforms

6.1 Development environment

Tool	Version	Usage
Kaggle Notebooks	Cloud	Main Python execution environment
Google Earth Engine	JavaScript API	Sentinel-2 and Sentinel-1 acquisition
Python	3.10+	Main language
Pandas	2.x	Tabular manipulation
GeoPandas	0.14+	Geospatial data (RPG, parcels)
Rasterio	1.3+	SoilGrids raster reading (WCS)
Requests	—	NASA POWER download (API)
OpenPyXL	—	Céré'Obs Excel reading
Matplotlib / Seaborn	—	EDA visualizations

6.2 External data sources

Source	Access platform	API type
Céré'Obs	Public Excel file (FranceAgriMer)	Direct download
RPG	data.cquest.org (mirror)	Shapefiles per region
NASA POWER	power.larc.nasa.gov	REST API JSON
SoilGrids	maps.isric.org	WCS (Web Coverage Service)
Sentinel-2	Google Earth Engine	JavaScript API (GEE)
Sentinel-1	Google Earth Engine	JavaScript API (GEE)

6.3 Deposit platforms

Platform	Usage	Main URL
Kaggle	Public deposit of 6 raw datasets	kaggle.com/rgislikassi
Google Drive	Raw outputs from Google Earth Engine	"GEE_Wheat_Thesis" folder

6.4 Reproducibility

Reproducibility commitment:

- All raw datasets are **public** on Kaggle (CC BY 4.0 license)
- Acquisition scripts are based on **free tools** (Kaggle, GEE, public APIs)
- The **random seed** is fixed (random_state=42) for RPG drawing
- Software **versions** are documented in Kaggle notebooks
- The entire **methodology is documented** in 19 public PDFs

Any researcher can, by following this report, reproduce the entire dataset.

PART III — Source-by-source acquisition

7. Céré'Obs acquisition (target)

Methodology for extracting the regional 50%-harvest date

7.1 Source identification

The **optimal harvest window** is the target information of the project. Without this reference data, no supervised learning is possible. Our research identified several potential candidates:

Candidate source	Spatial accuracy	Public availability	Verdict
Céré'Obs (FranceAgriMer)	Regional	✅ Public Excel	⭐ Chosen
Agreste (statistics)	Departmental	✅ Tables	Too late (post-harvest)
Sentinel-2 (detection)	Parcel-level	—	Tautological (input variable)
Farmer data	Parcel-level	❌ Private	Inaccessible

7.2 Source file download

The Céré'Obs Excel file is available on the FranceAgriMer website. Filename: `SCR-GRC-CEREOBS_CP_depuis_2015-A26.xlsx`.

> **Note:** this Excel file contains data for all cereals (soft wheat, durum wheat, barley, etc.) for all French regions since 2015. It is a gold mine for phenological research in France.

7.3 Extraction by linear interpolation

The **50%-harvest date** is not explicitly given. The file provides, week by week, the percentage of regional area harvested. Our extraction uses **linear interpolation**:

1. Filtering: CULTURE = "BTH" (Winter Soft Wheat) AND REGION = "Centre-Val de Loire"
2. For each year, identify the **two weeks bracketing 50%**:
    - Week N: SURFACE_RECOLTEE_PCT < 50%
    - Week N+1: SURFACE_RECOLTEE_PCT ≥ 50%
3. Linear interpolation between the two dates to find the exact day reaching 50%
4. Conversion to DOY (Day of Year)

7.4 Python code for extraction

```
import pandas as pd

df = pd.read_excel('SCR-GRC-CEREOBS_CP_depuis_2015-A26.xlsx')
df = df[(df['CULTURE'] == 'BTH') & (df['REGION'] == 'Centre-Val de Loire')]

results = []
for year, group in df.groupby('CAMPAGNE'):
    group = group.sort_values('DATE_DEBUT_SEMAINE')
    # Find the two weeks bracketing 50%
    before = group[group['SURFACE_RECOLTEE_PCT'] < 50].iloc[-1]
    after = group[group['SURFACE_RECOLTEE_PCT'] >= 50].iloc[0]
    # Linear interpolation
    pct_diff = after['SURFACE_RECOLTEE_PCT'] - before['SURFACE_RECOLTEE_PCT']
    date_diff = (after['DATE_DEBUT_SEMAINE'] - before['DATE_DEBUT_SEMAINE']).days
    days_to_50 = (50 - before['SURFACE_RECOLTEE_PCT']) / pct_diff * date_diff
    harvest_date = before['DATE_DEBUT_SEMAINE'] + pd.Timedelta(days=days_to_50)
    results.append({
        'year': year,
        'harvest_date': harvest_date,
```

```
        'harvest_doy': harvest_date.dayofyear
    })

target = pd.DataFrame(results)
```

7.5 Results obtained

Year	50%-harvest DOY	Calendar date	Particularities
2020	185	July 3, 2020	Average year, standard sunshine
2021	199	July 18, 2021	Cool spring slows maturation (+14j)
2022	183	July 2, 2022	May heatwave accelerates maturation (-2j)
2023	184	July 3, 2023	Average conditions
2024	195	July 13, 2024	Wet spring, delay (+10j)

The inter-annual variability is **16 days** between 2022 (earliest) and 2021 (latest). This variability is what our ML model will seek to explain.

7.6 Target limitations

Important limitation: the target is **regional**, not **parcel-level**. All parcels in the same year receive the same `harvest_doy` value.

Consequences:

- Our model predicts the **regional mean date**, not the individual date
- Parcel features (soil, S2, S1, weather) serve to explain **years**, not individual parcels
- Ideally, we would have per-parcel harvest dates, but such data is not public

Mitigation: we can hope that a well-trained model will capture parcel-level effects by exploiting correlations between features and regional targets. This is a research question in itself.

8. RPG acquisition (parcels)

Random selection of 1500 wheat parcels in CVL

8.1 Source: Graphic Parcel Registry

The **RPG** is the French national database listing all agricultural parcels declared by farmers as part of the CAP. It is published annually by IGN and the Ministry of Agriculture.

For license reasons, the official RPG is not directly downloadable. We used a **public mirror**: `data.cquest.org`, which distributes shapefiles by region and year under an open license.

8.2 Shapefile download

For each year (2020-2024), we downloaded the RPG shapefile for the **Centre-Val de Loire** region. Each shapefile contains ~80,000 to 90,000 soft wheat parcels:

Year	Total BTH parcels in CVL
2020	75,903
2021	89,171
2022	83,206
2023	80,346
2024	69,785

8.3 Sampling strategy

From this annual population, we performed **stratified random sampling**:

1. Filtering: `surf_parc >= 1.0` hectare (elimination of micro-parcels)
2. Random draw: 300 parcels per year
3. Seed: `random_state=42` (for reproducibility)
4. WGS84 centroid computation (for subsequent API queries)

8.4 Validation of temporal independence

An important question: should we sample the **same 300 parcels** each year (temporal follow-up), or **300 new** ones each year (independent samples)?

> **Analysis conducted:** we calculated the **crop rotation** rate in CVL. Over 2 consecutive years, only **17.1%** of parcels carry wheat. This is consistent with agronomic practices (rotation every 2-3 years).

Consequence: following the same parcels would have drastically reduced our sample (from 300 to ~50 in common over 5 years). We therefore chose **independent annual sampling**, which:

- Maximizes sample size (300 × 5 = 1500)
- Captures more agronomic diversity (different soils, farmers)
- Avoids continuity biases ("good" parcels carrying wheat every year may not be representative)

8.5 Final schema

The final file `parcelles_ble_cvl_sample.csv` contains 1500 rows with the following columns:

Column	Source	Example
parcelle_uid	Constructed (year_idparcel)	2020_4771311
year	Constructed	2020

id_parcel	Original RPG	4771311
code_cultu	RPG	BTH
surf_parc	RPG (m^2) → ha	5.2
departement	INSEE code	28
commune	INSEE code	28085
latitude, longitude	WGS84 centroid	48.4523, 1.4912

8.6 Visual validation

> **Control performed:** we generated a map of the 1500 parcels and visually verified that they are well distributed across the 6 CVL departments, with a density consistent with agricultural reality (denser in Beauce, less dense in Sologne and Boischaut).

9. NASA POWER acquisition (weather)

Daily climate variable download for 6 CVL points

9.1 Choice of NASA POWER service

Several weather sources were candidates:

Source	Spatial resolution	Access	Verdict
Météo-France OpenData	Field stations (~5 km)	Limited API	Variable coverage, API complications
ECMWF ERA5	0.25° (~28 km)	Climate Data Store	Excellent but complex to integrate
NASA POWER	0.5° (~55 km)	Simple REST API	⭐ Chosen (simplicity + sufficient)
ECMWF MARS	Very fine	Academic	Not open data

9.2 Extraction points

NASA POWER provides data for any lat/lon point. For our project, we extract **6 points** corresponding to the prefectures of the 6 CVL departments:

Department	Prefecture	Latitude	Longitude
28	Chartres	48.45°N	1.49°E
45	Orléans	47.90°N	1.91°E
41	Blois	47.59°N	1.33°E
37	Tours	47.39°N	0.69°E
18	Bourges	47.08°N	2.40°E
36	Châteauroux	46.81°N	1.69°E

Each parcel is associated with the weather of its departmental prefecture (join by `departement`). This level of granularity is consistent with the native resolution of NASA POWER (0.5° = ~55 km).

9.3 Period and downloaded variables

- **Period:** 2019-10-01 → 2024-07-31 (1,766 days)
- **Raw variables:** 8 (T2M, T2M_MIN, T2M_MAX, PRECTOTCORR, RH2M, WS2M, ALLSKY_SFC_SW_DWN, PS)
- **Return format:** JSON via REST API

9.4 Computed derived variables

From the 8 raw variables, we computed **22 agronomic indicators** (detailed in the Data Dictionary, Section 7.4). The most important are:

Category	Key variables	Agronomic interest
Growing degree-days	gdd_base0_cumul, gdd_base5_cumul	Cumulative maturation indicator
Evapotranspiration	et0, et0_cumul	Atmospheric water demand
Water balance	bilan_hydrique, bilan_hydrique_cumul	Cumulative water stress
Thermal stress	jours_chauds, amplitude_therm	Heatwave risk, grain shrivelling
Seasonal cumulations	pluie_30j, rayonnement_cumul	Resource availability

9.5 Critical bug identified and corrected: ET0

Initial bug: in the first version of the pipeline, we mistakenly applied a `× 3.6` conversion to the ALLSKY_SFC_SW_DWN variable before computing ET0, assuming it was provided in Wh/m². In reality, NASA POWER provides this variable directly in **MJ/m²/day**.

Consequence of the bug: the calculated ET0 values were artificially multiplied by ~3.6, giving unrealistic ET0 (15-20 mm/day instead of typical 4-6 mm/day).

Correction: we removed the incorrect conversion. The current version uses native NASA POWER values without transformation.

Post-correction validation: mean ET0 over June-July ≈ 4.5 mm/day, consistent with FAO reference values for metropolitan France.

9.6 Validation of obtained values

We validated values by comparison to CVL agronomic references:

Variable	Observed value (mean)	Agronomic reference	Verdict
Annual mean T	11.5 °C	~11 °C (Météo-France)	✅
Annual rainfall	650 mm	600-700 mm (CVL)	✅
Annual ET0	1,000 mm	950-1,050 mm (FAO)	✅
GDD base 0 cumul to June	1,800-2,200 °C·j	~2,000 °C·j	✅
Rainfall cumul Oct-Jul	520 mm	500-600 mm	✅

10. SoilGrids acquisition (soil)

Extraction of 39 features via WCS rasters (91× optimization)

10.1 Choice of SoilGrids

Several soil data sources were candidates:

Source	Resolution	Coverage	Verdict
BDAT (INRAE France)	Very precise but punctual	Field samples	No continuous coverage
RMQS (INRAE)	Excellent	~2,000 sites in France	Too punctual for 1500 parcels
SoilGrids	250 m, predictive ML	Worldwide	⭐ Chosen (coverage + resolution)
European Soil Data Centre	1 km	Europe	Less precise than SoilGrids

10.2 First attempt: REST API

SoilGrids provides a REST API allowing value extraction at a given lat/lon point. First approach attempted:

```
# Pseudo-code of initial approach (REST API)
for parcel in 1500_parcels:
    for property in 13_properties:
        for depth in 3_depths:
            url = f"https://rest.isric.org/soilgrids/v2.0/properties/query?"
            url += f"lon={parcel.lon}&lat={parcel.lat}&property={property}"
            url += f"&depth={depth}"
            response = requests.get(url)
            # Store the value

# Estimate: 1500 × 13 × 3 = 58,500 requests
# Estimated time (rate-limited): ~5 hours
```

10.3 Optimization: WCS raster extraction

Methodological innovation: instead of 58,500 individual API requests, we download **39 complete rasters** (Web Coverage Service) covering all of CVL, then sample the 1500 points locally.

```
# Optimized WCS approach
import rasterio
from owslib.wcs import WebCoverageService

wcs = WebCoverageService('https://maps.isric.org/mapserv?map=/map/SoilGrids')
for property in 13_properties:
    for depth in 3_depths:
        # Download 1 raster for the CVL zone
        raster = wcs.getCoverage(
            identifier=f"{property}_{depth}_mean",
            bbox=cvl_bounds, crs='EPSG:4326', format='GEOTIFF'
        )
        # Sample the 1500 points locally
        with rasterio.open(raster) as src:
            for parcel in 1500_parcels:
                values[parcel.uid] = src.sample([(parcel.lon, parcel.lat)])

# 39 downloads × ~3 seconds + local sampling
# Measured time: 3.3 minutes (vs 5 hours REST API)
# Gain: 91× faster
```

10.4 Handling missing values

Out of 1500 parcels, **2 (0.13%)** had NoData values for certain properties, due to SoilGrids raster artifacts near administrative borders.

We applied **nearest-neighbor imputation**:

1. Identify parcels with NoData
2. Find the neighboring parcel (in Euclidean distance) with a valid value
3. Copy the neighbor's value

> **Agronomic justification:** soil properties vary little over short distances (a few hundred meters). The mean distance to nearest neighbor was **0.22 km** (= 1 SoilGrids pixel), so imputation is physically reasonable.

10.5 Value validation

Property	Observed CVL median	INRAE reference	Verdict
Clay (g/kg, 0-30 cm)	185	150-200 (CVL)	✅
Silt (g/kg)	420	400-500 (Beauce)	✅
Sand (g/kg)	395	300-450	✅
pH water	6.7	6.5-7.0 (typical wheat)	✅
SOC (dg/kg)	148	100-200	✅
CEC (cmol+/kg)	15.2	10-25	✅

All observed medians fall within INRAE reference ranges for CVL, validating the quality of SoilGrids extraction.

11. Sentinel-2 acquisition (optical)

3 temporal granularities · 1500 parcels · 5 years

11.1 Platform choice: Google Earth Engine

For Sentinel-2, several access platforms were candidates:

Platform	Advantages	Disadvantages	Verdict
Copernicus Hub	Official ESA source	GB downloads, complex local processing	Unsuitable for 1500 parcels
SentinelHub	Ergonomic API	Paid beyond a quota	Prohibitive cost
Google Earth Engine	Free cloud computing, COPERNICUS/S2_SR_HARMONIZED ready	Specific JavaScript API	⭐ Chosen
AWS Open Data	Raw S2 accessible	No pre-processing, complexity	Too low-level

11.2 Uploading parcels asset to GEE

First step: upload our 1500-parcel sample as a **GEE asset** to use in JavaScript scripts:

1. Conversion of `parcelles_ble_cvl_sample.geojson` to GEE-compatible shapefile format
2. Manual upload in GEE console: `projects/gen-lang-client-0405586640/assets/parcelles_ble_cvl_sample`
3. Verification: 1500 features with attributes (parcelle_uid, year, departement, etc.)

11.3 GEE pipeline — 15-day composites

The JavaScript pipeline to produce 15-day composites is summarized as follows:

```
var parcelles = ee.FeatureCollection(
  'projects/gen-lang-client-0405586640/assets/parcelles_ble_cvl_sample'
);

// Load harmonized collection
var s2 = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
  .filterDate('2019-10-01', '2024-07-31')
  .filterBounds(parcelles)
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 60));

// Cloud masking via SCL (Scene Classification Layer)
function maskClouds(image) {
  var scl = image.select('SCL');
  var mask = scl.neq(3).and(scl.neq(7)).and(scl.neq(8))
              .and(scl.neq(9)).and(scl.neq(10)).and(scl.neq(11));
  return image.updateMask(mask);
}

// Compute NDVI, EVI, NDWI
function computeIndices(image) {
  var ndvi = image.normalizedDifference(['B8', 'B4']).rename('NDVI');
  var ndwi = image.normalizedDifference(['B8', 'B11']).rename('NDWI');
  var evi = image.expression(
    '2.5 * (NIR - RED) / (NIR + 6 * RED - 7.5 * BLUE + 1)', {
      'NIR': image.select('B8').divide(10000),
      'RED': image.select('B4').divide(10000),
      'BLUE': image.select('B2').divide(10000)
    }).rename('EVI');
  return image.addBands([ndvi, evi, ndwi]);
}

// 15-day composites over the window 2019-10 → 2024-07
var windowStart = ee.Date('2019-10-01');
var nWindows = 118; // 118 windows of 15 days
var windows = ee.List.sequence(0, nWindows - 1).map(function(i) {
```

```
  var start = windowStart.advance(ee.Number(i).multiply(15), 'day');
  var end = start.advance(15, 'day');
  var subset = s2.filterDate(start, end).map(maskClouds).map(computeIndices);

  // Empty window check
  var count = subset.size();
  var composite = ee.Algorithms.If(
    count.gt(0),
    subset.select(['NDVI', 'EVI', 'NDWI']).reduce(
      ee.Reducer.median().combine(ee.Reducer.mean(), '', true)
        .combine(ee.Reducer.minMax(), '', true)
        .combine(ee.Reducer.stdDev(), '', true)
    ).set({'window_start': start, 'n_images': count}),
    null
  );
  return composite;
});

// reduceRegions per parcel, CSV export
var results = windows.filter(ee.Filter.notNull(['window_start']))
  .map(function(image) {
    return ee.Image(image).reduceRegions({
      collection: parcelles,
      reducer: ee.Reducer.mean(),
      scale: 20
    });
  });

Export.table.toDrive({
  collection: ee.FeatureCollection(results).flatten(),
  description: 's2_composites_15days_cvl',
  fileFormat: 'CSV'
});
```

11.4 Empty window handling (critical bug avoided)

> **Problem encountered:** on some 15-day windows, no valid Sentinel-2 scene was available (very cloudy winter, satellite maintenance, etc.). Without precaution, GEE crashes with an "Empty collection" error.
>
> **Implemented solution:** use of `ee.Algorithms.If(count.gt(0), ..., null)` to dynamically handle empty collection cases. Empty windows return `null` and are filtered by `ee.Filter.notNull(['window_start'])`.
>
> **Result:** the pipeline never crashes, even in the presence of large coverage gaps.

11.5 The 3 granularities produced

The same pipeline is executed 3 times with different parameters to produce **3 files**:

Granularity	File	Volume	Intended use
15 days	s2_composites_15days_cvl.csv	~42 MB, 177k rows	Classical ML (RF, XGBoost)
7 days	s2_composites_7days_cvl.csv	~85 MB, ~380k rows	Sequential models (LSTM, GRU)
Raw per scene (5 files/year)	sentinel2_raw_{2020-2024}.csv	~40 MB × 5 = 200 MB	Transformer, maximum flexibility

11.6 Validation of obtained data

Seasonal NDVI distribution

To validate quality, we analyzed NDVI distribution month by month:

Month	Observed median NDVI	Expected NDVI (wheat)	Verdict
November (emergence)	0.32	0.30-0.40	✅
February (tillering)	0.41	0.40-0.50	✅

April (stem elongation)	0.68	0.60-0.75	✅
May (heading/peak)	0.84	0.80-0.90	✅
June (maturation)	0.58	0.50-0.70	✅
July (post-harvest)	0.22	0.15-0.30	✅

NDVI peak across years

Year	NDVI peak (DOY)	Calendar date	Consistency with harvest (DOY)
2020	134	May 13	Harvest 185 → 51 d after peak ✅
2021	143	May 23	Harvest 199 → 56 d after peak ✅
2022	123	May 3	Harvest 183 → 60 d after peak ✅
2023	118	April 28	Harvest 184 → 66 d after peak ⚠️
2024	128	May 7	Harvest 195 → 67 d after peak ⚠️

> **Lesson learned:** the gap between NDVI peak and harvest varies between 51 and 67 days depending on the year. This is consistent with the fact that the peak → harvest phase depends strongly on temperature (hot summer accelerates, cool summer slows). This variable gap is precisely the information our ML model will seek to exploit.

11.7 Identified limitations

> **Sentinel-2 limitations in our project:**
>
> - **Degraded winter coverage:** 50-70% of invalid scenes in December-February due to clouds
> - **EVI outliers:** 0.12% of EVI measurements are aberrant (to filter)
> - **NDVI ↔ NDWI highly correlated** (r = 0.93), creating multicollinearity for some models
> - **Edge effects** on small parcels: parcel average includes some mixed peripheral pixels

12. Sentinel-1 SAR acquisition (radar)

All-weather VV/VH backscatter · 5 annual files

12.1 Motivation for including Sentinel-1

Adding Sentinel-1 SAR to Sentinel-2 addresses several optical limitations:

S2 limitation	S1 solution
50-70% clouds in winter	SAR passes through clouds → ~8 scenes/month in winter too
Reflectance measurement (passive)	3D structure measurement (active)
NDVI saturation at 0.85+	No saturation, biomass-sensitive until harvest
Harvest signal = NDVI drop	Harvest signal = sharp VV/VH ratio rise

12.2 Chosen configuration

Several configuration parameters were possible. Our choice:

Parameter	Choice	Rationale
Collection	COPERNICUS/S1_GRD	Ground Range Detected, calibrated in dB
Mode	IW (Interferometric Wide)	Standard for terrestrial observation, 250 km swath
Polarisations	VV + VH	Transmitted polarisations on land
Orbit pass	DESCENDING only	Consistent geometry, stable incidence angle
Final resolution	30 m	Oversampling from 10m, sufficient for parcels
Speckle filter	focal_median 50 m	Reduces SAR granular noise

> **Why DESCENDING only?** By combining ASCENDING and DESCENDING, we get more scenes but with different geometries (incidence angles ~30° vs ~40°), creating discontinuities in the time series. By limiting to DESCENDING, we lose ~50% of available scenes but gain radiometric consistency.

12.3 GEE pipeline — VV/VH backscatter

```
var parcelles = ee.FeatureCollection(
  'projects/gen-lang-client-0405586640/assets/parcelles_ble_cvl_sample'
);

// Load the collection
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
  .filterDate('2019-10-01', '2024-07-31')
  .filterBounds(parcelles)
  .filter(ee.Filter.eq('instrumentMode', 'IW'))
  .filter(ee.Filter.eq('orbitProperties_pass', 'DESCENDING'))
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VH'));

// Speckle filter + VV/VH ratio
function processImage(image) {
  // Local median speckle filter 50m
  var vv = image.select('VV').focal_median(50, 'circle', 'meters');
  var vh = image.select('VH').focal_median(50, 'circle', 'meters');

  // VV/VH ratio in dB (log difference = ratio)
  var ratio = vv.subtract(vh).rename('VV_VH_ratio');

  return image.addBands([
```

```
    vv.rename('VV_filtered'),
    vh.rename('VH_filtered'),
    ratio
  ]).set('system:time_start', image.get('system:time_start'));
}

var processed = s1.map(processImage);

// For each scene, extract per-parcel means
var scenesToFeatures = processed.map(function(image) {
  return image.reduceRegions({
    collection: parcelles,
    reducer: ee.Reducer.mean(),
    scale: 30
  }).map(function(f) {
    return f.set('scene_date', ee.Date(image.get('system:time_start'))
      .format('YYYY-MM-dd'));
  });
});

// Export per year
[2020, 2021, 2022, 2023, 2024].forEach(function(year) {
  var yearStart = ee.Date.fromYMD(year - 1, 10, 1);
  var yearEnd = ee.Date.fromYMD(year, 7, 31);
  var yearScenes = processed.filterDate(yearStart, yearEnd);
  // ... export to drive
});
```

12.4 Export structure

The pipeline produces **5 files** (one per agricultural season):

File	Period covered	Approximate size
sentinel1_sar_raw_2020.csv	2019-10-01 → 2020-07-31	~25 MB
sentinel1_sar_raw_2021.csv	2020-10-01 → 2021-07-31	~25 MB
sentinel1_sar_raw_2022.csv	2021-10-01 → 2022-07-31	~25 MB
sentinel1_sar_raw_2023.csv	2022-10-01 → 2023-07-31	~25 MB
sentinel1_sar_raw_2024.csv	2023-10-01 → 2024-07-31	~25 MB

12.5 Validation and phenological signature

To validate SAR data quality, we analyzed the phenological signature over the 2020 season:

Period	VV median (dB)	VH median (dB)	Ratio (dB)	Phenological stage
Oct 2019	-9.2	-17.2	8.0	Sowing (rough bare soil)
Nov-Dec	-8.9	-17.5	8.6	Emergence (low canopy)
Jan-Feb 2020	-9.0	-18.1	9.1	Tillering (slow growth)
March	-10.7	-18.3	7.6	Stem elongation begins
Apr-May	**-13.5**	**-19.1**	**5.6**	**Heading/flowering (max canopy)**
June	-12.3	-18.7	6.4	Maturation (senescence)
July (post-harvest)	-9.8	-17.6	7.8	Harvest → rebound

Perfect phenological signature: VV values decrease from -9 dB (bare soil) to -13.5 dB (canopy peak) then return to -9.8 dB (post-harvest). This "inverted U" signature is the expected agronomic reference for wheat. **SAR is working as expected.**

12.6 Coverage statistics

VALID SCENES / PARCEL (MEDIAN)	MAX SCENES / PARCEL	S1 DENSITY / MONTH	VS SENTINEL-2
100	**124**	**~8**	**2.4×**

12.7 Measured S1+S2 complementarity

The main benefit of SAR is filling in Sentinel-2 cloud gaps. We measured this complementarity:

Month	S2 scenes/parc	S1 scenes/parc	Combined total
December	2.6	8.0	10.6
January	5.2	8.0	13.2
February	2.4	8.0	10.4
March	4.2	8.0	12.2
April	8.0	8.0	16.0
May	9.6	8.0	17.6
June	10.2	8.0	18.2
July	10.8	8.0	18.8
Mean	**6.6**	**8.0**	**14.6**

Combined S1+S2 density reaches **~15 scenes/parcel/month on average**, with a substantial portion in winter (when S2 alone provided only 2-5 scenes). This combination makes our dataset one of the densest publicly available for wheat monitoring.

PART IV — Methodological lessons learned

This part capitalizes on the technical and methodological lessons learned from several months of work. It aims to serve as **feedback** for future students or researchers working on similar projects.

13. Bugs identified and fixed

13.1 Bug #1 — Erroneous radiation conversion (ET0)

Symptom: ET0 values calculated by our NASA POWER pipeline were abnormally high (15-20 mm/day in summer), vs 4-6 mm/day expected.

Root cause: we had applied a `× 3.6` conversion to the ALLSKY_SFC_SW_DWN variable, assuming it was provided in Wh/m² (in which case conversion to MJ/m²/day would be necessary). In reality, NASA POWER delivers this variable directly in **MJ/m²/day**.

Consequence: daily ET0 was multiplied by ~3.6, and seasonal cumulations were totally unrealistic.

Correction: removal of incorrect conversion. Post-correction validation by comparison to FAO references.

Lesson learned: *always check native units of external APIs before manipulating variables, even if the documentation seems clear.* For the future, add a systematic "unit validation" phase in all pipelines.

13.2 Bug #2 — GEE crash on empty time windows

Symptom: the Sentinel-2 pipeline crashed when processing some 15-day windows, without an explicit error message, in the Google Earth Engine console.

Root cause: some windows contained no valid Sentinel-2 scene (very cloudy winter, satellite maintenance). The call to `imageCollection.reduce(...)` on an empty collection generated a fatal error.

Correction: implementation of dynamic verification with `ee.Algorithms.If(count.gt(0), composite, null)`, then filtering nulls downstream.

Lesson learned: *distributed (cloud) pipelines do not tolerate degenerate cases. We must always code defensively, especially on long temporal sources where gaps are inevitable.*

13.3 Bug #3 — Undetected SoilGrids NoData values

Symptom: some parcels had aberrant values for soil (pH = 0, clay = 0).

Root cause: SoilGrids encodes NoData values differently depending on the property (sometimes 0, sometimes -32768, sometimes NaN). Our initial rasterio reading didn't recognize all cases.

Correction:

1. Explicit reading of the `nodata` attribute of each raster
2. Systematic conversion to NaN before any calculation
3. Nearest-neighbor imputation for the 2 concerned parcels

Lesson learned: *NoData conventions vary between sources and between properties. Always verify explicitly (and don't trust apparent values).*

13.4 Bug #4 — Edge effects on 7-day composites

Symptom: 7-day composites sometimes showed abnormal "dips" mid-season, while 15-day composites over the same period were clean.

Root cause: a 7-day window may contain 0 or 1 valid scenes (whereas a 15-day window has 2-3). When there is only one scene, and it is partially cloudy, the extracted NDVI is biased by masked pixels.

Correction: documentation of this limitation. For sequential models (LSTM, Transformer) using 7-day composites, recommend post-hoc filtering `n_images >= 2` in pre-processing.

Lesson learned: *the finer the granularity, the more noise-sensitive. We must document these trade-offs for downstream users.*

13.5 Bug summary

Bug	Source	Severity	Status
Conversion ET0 ×3.6	NASA POWER	Critical	✅ Fixed
Empty windows crash	Sentinel-2 GEE	Critical	✅ Fixed
NoData SoilGrids	SoilGrids	Medium	✅ Fixed (imputation)
7-day composite noise	Sentinel-2	Low	📝 Documented

14. Major optimizations

14.1 Optimization #1 — SoilGrids via WCS (91× faster)

Initial problem: the SoilGrids REST API allowed extracting the 39 properties (13 × 3 depths) point by point. For 1500 parcels, this meant 58,500 API requests, i.e., ~5 hours of downloading (rate-limited).

Implemented solution: download the 39 complete rasters via WCS for the CVL area, then sample the 1500 points locally with rasterio.

Measured gain: 3.3 minutes (vs 5 hours), a factor **91× faster**.

Bonus: the downloaded rasters can be reused to extract additional parcels without new downloading.

Lesson learned: *faced with a rate-limited REST API, always evaluate if a bulk alternative exists (WCS, FTP, dump). The gain can be colossal.*

14.2 Optimization #2 — GEE composites rather than raw scenes

Initial problem: exporting each Sentinel-2 scene individually would have produced ~1700 rows per parcel × 1500 parcels = 2.5 million rows per year, i.e., ~200 MB × 5 years = 1 GB. The size made downstream (Pandas) manipulation slow.

Implemented solution: produce 15-day and 7-day composites **within GEE**, before export. This reduces volume while preserving relevant temporal information for classical ML.

Measured gain: 42 MB (15j composites) vs 200 MB (raw), **5× lighter**, without loss of information useful for tabular models.

Assumed trade-off: we also kept raw (5 files of 40 MB) for architectures requiring maximum temporal resolution (Transformer).

Lesson learned: *aggregation can be done at different pipeline levels (upstream in GEE, downstream in Pandas). Doing it upstream when the final usage is well known saves bandwidth and processing time.*

14.3 Optimization #3 — Quality filter at the collection level

Initial problem: without pre-filtering, GEE processes thousands of scenes per window, many of which are almost completely cloudy.

Implemented solution: apply a filter `CLOUDY_PIXEL_PERCENTAGE < 60%` from the `filterMetadata` step, before any computation.

Measured gain: ~30% fewer scenes to process, so 30% less time. The 60% threshold is a trade-off: more restrictive (40%) you lose entire windows in winter; more permissive (80%) you keep unusable scenes.

Lesson learned: *filter as early as possible in the pipeline. GEE's "lazy computations" allow this filtering for free.*

14.4 Optimization #4 — Documentation parallel to datasets

Initial problem: at the start of the project, documentation was done "at the end", which created omissions (undocumented variables, forgotten methods).

Implemented solution: as soon as a dataset is finalized, produce **simultaneously** a Data Dictionary and a Dataset Report PDF. Each Kaggle deposit is therefore accompanied by immediate documentation.

Gain: always up-to-date documentation, no rework at the end of the project, estimated time savings of 1-2 days over the whole thesis.

Lesson learned: *"document as you go" — running documentation is less costly than retroactive final documentation.*

15. Decisions and trade-offs

Beyond bugs and optimizations, several **methodological decisions** were made that deserve to be made explicit with their trade-offs.

15.1 Regional vs national selection

Option	Advantages	Disadvantages
1 region (CVL) — **retained**	Manageable pipeline, homogeneous climate	Generalization to other regions to verify
3 regions	More climate diversity	3× more data volume, increased complexity
Entire France	More generalizable model	Massive GEE pipeline, edge cases (Mediterranean, mountain)

The choice of a single region (CVL) is a trade-off between **feasibility within a master's thesis** and **sufficient agronomic diversity**. The methodology is designed to be **scalable**: we can extend to other regions by relaunching the pipeline.

15.2 Sample size (1500 parcels)

Sample	Advantages	Disadvantages
500 parcels	Fast pipeline (1-2h GEE)	Low statistical power
1500 (300/year × 5) — **retained**	Speed/diversity trade-off, viable tabular ML	Limited for massive deep learning
10,000 parcels	Ideal for DL	Multi-day GEE pipeline, Drive quotas

1500 parcels is sufficient to train tabular models (RF, XGBoost) with cross-validation. For heavy deep learning architectures (TFT, multimodal), it's borderline but exploitable with anti-overfitting strategies (regularization, dropout, early stopping).

15.3 Period 2020-2024 vs longer

Period	Advantages	Disadvantages
2015-2024 (10 years)	+ climate variability	Sentinel-2 starts 2015, S1 starts 2014, limited Céré'Obs data before
2020-2024 (5 years) — **retained**	All sources consistent, S2_SR_HARMONIZED available	Limited temporal sample
2023-2024 (2 years)	Very recent data	Not enough for ML

5 years is the minimum to have significant climate variability (2020 average, 2021 cool, 2022 hot, 2023 average, 2024 wet). It's also the period where all our sources are aligned in quality.

15.4 Modalities included vs excluded

Modality considered	Status	Justification
NDVI/EVI/NDWI (Sentinel-2)	✅ Included	Standard of agricultural remote sensing
VV/VH (Sentinel-1)	✅ Included	Complements optical, all-weather
Soil (SoilGrids)	✅ Included	Essential explanatory variable
Weather (NASA POWER)	✅ Included	Main driver of phenology
Topography (SRTM)	❌ Excluded	Very flat CVL, low altitude variability
MODIS (250 m)	❌ Excluded	Insufficient resolution for parcels
Farmer data	❌ Excluded	Private, not scalable
Paid satellite data (Planet)	❌ Excluded	Cost, not open data

15.5 Regional vs parcel-level target

Major project trade-off: the target (50%-harvest date) is **regional**, creating an asymmetry: we have 1500 input observations (parcel-level), but only 5 distinct targets (one per year).

Limitations:

- The model learns yearly patterns, not really parcel patterns
- Cross-validation evaluation must be done **by year** (group k-fold), not random split
- With 5 targets, we are limited to 5-fold leave-one-year-out validation

Benefits despite this:

- Parcel features capture the **spatial heterogeneity** that modulates timing
- A parcel in deep silty soil and one in sandy soil don't have the same optimal date even in the same year
- Aggregation to the regional target smooths this variation, but it remains partially detectable

Future avenues: obtain parcel-level harvest dates via partnerships with agricultural cooperatives (Dijon Céréales, Axéréal, etc.) or via automatic S1/S2 detection to generate a synthetic parcel-level target.

PART V — Validation and quality

16. Validations performed

This section summarizes all validations conducted on the acquired dataset.

16.1 Target validation (Céré'Obs)

Check	Method	Result
Date range	183 ≤ DOY ≤ 199	✅ Consistent with winter wheat calendar
Inter-annual variability	Max - min spread = 16 days	✅ Consistent with literature (10-20j)
Climate consistency	2021 (cool spring) = delay, 2022 (May heatwave) = early	✅ Agronomic logic
Historical comparison	2020-2024 harvest aligned with Agreste	✅ Concordance

16.2 RPG validation

Check	Method	Result
Temporal independence	Crop rotation calculation	17.1% persistence ✅ Good
Min area respected	min(surf_parc) ≥ 1.0	✅ 1.0 ha
Department coverage	Histogram by dpt	✅ 6 departments represented
Valid WGS84 coordinates	Lat/lon extent	✅ Within CVL bbox
Reproducibility	Re-draw with seed=42	✅ Identical sample

16.3 NASA POWER validation

Check	Method	Result
Annual mean T	Météo-France comparison	11.5°C vs ~11°C ✅
Rainfall	Normals 1981-2010 comparison	650 mm/year ✅
ET0 (post-correction)	FAO comparison	1000 mm/year ✅
GDD cumul to end June	Arvalis reference	~2000 °C·j ✅
Inter-department consistency	Similar seasonal profiles	✅ Variations < 10%

16.4 SoilGrids validation

Check	Method	Result
Texture (clay/silt/sand)	INRAE-BDAT comparison	Consistent ranges ✅
pH	5.5-7.5 range expected for wheat	Median 6.7 ✅
Depth profiles	Consistent variations 0-5 → 15-30	✅
NoData handled	Neighbor imputation	✅ 2 parcels, distance < 0.5 km

16.5 Sentinel-2 validation

Check	Method	Result
NDVI ranges per stage	Phenology comparison table	All valid ✅

NDVI peak	0.80 - 0.90 expected in May	0.84 median ✅
EVI outliers	Filter |EVI| < 1.5	0.12% outliers ✅ Minor
NaN pattern	Concentrated December-February	✅ Consistent with winter clouds
NDVI/NDWI correlation	r = 0.93	⚠️ Known multicollinearity
Year consistency	Similar seasonal profiles	✅ With expected annual shifts

16.6 Sentinel-1 SAR validation

Check	Method	Result
VV ranges (-15 to -5 dB)	Median -10.7 dB	✅ Wheat range
VH ranges (-22 to -12 dB)	Median -18.5 dB	✅ Wheat range
VH < VV	Positive difference	✅ Always
Inverted U signature	VV: bare soil → canopy → bare soil	✅ Consistent
Post-harvest rebound	VV/VH ratio +1.5 dB July	✅ Visible
Scenes/parcel density	Median 100, max 124	✅ 2.4× Sentinel-2

17. Dataset quality indicators

17.1 Global dashboard

PARCELS	SOURCES	KAGGLE DATASETS	TOTAL VOLUME
1,500	**6 / 6**	**6 / 6**	**~620 MB**

17.2 Indicators per source

Source	Completeness	Agronomic validation	Tests performed
Céré'Obs	5/5 (100%)	✅ Ranges, variability	4
RPG	1500/1500 (100%)	✅ Validated sampling	5
NASA POWER	10596/10596 (100%)	✅ All consistent values	5
SoilGrids	1500/1500 after imputation (100%)	✅ INRAE-consistent textures	4
Sentinel-2	74% on 15j composites	✅ Valid NDVI profiles	6
Sentinel-1	20% structural (footprint)	✅ Valid SAR signature	6

17.3 Reusability indicator

FAIR criterion	Status	Implementation
Findable	✅	Kaggle datasets indexed by Google, keywords
Accessible	✅	Direct download on Kaggle, free
Interoperable	✅	Standard CSV/GeoJSON, UTF-8, WGS84
Reusable	✅	CC BY 4.0, 19 PDF documentation

18. Known limitations of the dataset

For honest and informed use of the dataset, here are the identified limitations:

18.1 Structural limitations

L1 — Regional target, not parcel-level: the 1500 parcels share only 5 distinct target values (one per year). This limits the model's ability to predict at fine parcel scale. *Mitigation:* parcel features nevertheless provide information about heterogeneity.

L2 — Sample limited to 5 years: insufficient to capture long-term climate variability. An extreme year (2003 heatwave, 2018 drought) is not represented. *Mitigation:* the period 2020-2024 still includes 5 climatically distinct years.

L3 — Limited geographic coverage (CVL): trained models will not automatically generalize to other regions (Mediterranean, North Atlantic, etc.). *Mitigation:* the methodology is scalable, the pipeline can be reused.

L4 — No farmer data: variety, sowing date, fertilization absent. This information would explain part of the remaining parcel variability. *Mitigation:* methodological choice assumed for reproducibility and scalability.

18.2 Technical limitations

L5 — Degraded winter S2 coverage: 50-70% of invalid scenes in December-February. *Mitigation:* S1 SAR fills the gaps.

L6 — NASA POWER resolution (0.5°): weather is attributed by department, not by parcel. *Mitigation:* in a homogeneous region like CVL, the error is limited. For future extension, consider ERA5 (0.25°) or Météo-France stations (~5 km).

L7 — SoilGrids resolution (250m): intra-parcel soil variations not captured. *Mitigation:* for parcels of 1+ ha, 250m is an acceptable resolution.

L8 — EVI composite can diverge: 0.12% of outliers to filter manually. *Mitigation:* documented, filter |EVI| < 1.5 recommended.

L9 — NDVI/NDWI correlation (r=0.93): information redundancy. *Mitigation:* problematic for linear models, not for trees or NN. Use PCA or feature selection if needed.

18.3 Upcoming limitations (ML phase)

- 1500 observations is borderline for heavy deep learning (TFT). Overfitting risk to anticipate.
- Strict cross-validation by year (leave-one-year-out) → only 5 folds.
- Result interpretation will be constrained by the regional nature of the target.

PART VI — Next steps

19. Master dataset construction

19.1 Step 1 — Derived features computation

From raw time series, compute the ~150 derived features defined in the Master Data Dictionary specification:

Family	Number of features	Estimated effort
Sentinel-2 (peak_doy, senescence_doy, integrals, monthly statistics)	~25	1-2 days
Sentinel-1 (vv_min_doy, ratio_recovery_doy, amplitudes)	~25	1-2 days
Weather (GDD at several dates, AMJ cumulations, hot days)	~40	1 day
Soil (already static, 39 SoilGrids features)	39	—
Metadata and quality flags	~10	0.5 day
TOTAL features	**~140**	**4-5 days**

19.2 Step 2 — Final fusion

Merge all feature blocks into a unique master dataset:

1. Load the 6 raw sources
2. Compute derived features per block (pandas, numpy)
3. Join by `parcelle_uid` (soil, S2, S1, weather, target)
4. Verify 1500 rows × ~150 columns
5. Export in CSV and Parquet
6. Publish as 7th Kaggle dataset: **"Master Dataset"**

19.3 Step 3 — Master documentation

Produce two final documents:

- **Master Dataset Data Dictionary v2**: with all finalized derived features
- **Master Dataset Report v2**: extended version of this document, integrating phases 4 + 5

20. Planned ML approach

20.1 Modeling strategy

Several model families will be compared, with a **progressive complexity** logic:

Phase	Models	Objective
P1 — Baseline	Mean predictor, linear regression	Establish performance floor
P2 — Classical ML	Random Forest, XGBoost, LightGBM	Interpretable tabular models, solid baseline
P3 — Sequential	LSTM, GRU with raw S2/S1 features	Exploiting time series
P4 — Advanced	Temporal Fusion Transformer (TFT)	State-of-the-art for multivariate series
P5 — Multimodal	Custom architecture fusing S2 + S1 + tabular	Fully exploit multi-source richness

20.2 Evaluation strategy

> **Cross-validation: Leave-One-Year-Out (LOYO)** with the 5 years as folds.
> - Train on 4 years, test on 1 year
> - Repeated 5 times
> - Metrics: RMSE, MAE, R^2 on predicted harvest DOY
>
> **Justification:** avoids temporal leakage. It is also the real scenario of operational use.

20.3 Interpretability

Beyond pure performance, the analysis of **feature importance** is crucial:

- **SHAP values** for XGBoost and Random Forest
- **Attention weights** for Transformer (TFT)
- **Permutation importance** to identify key features
- **Partial dependence plots** to understand non-linear effects

This will help answer the research questions: "Which signals are most predictive? S2? S1? Soil? Weather?"

21. Conclusion and perspectives

21.1 What has been achieved

At this stage of the project (end of acquisition phase):

- ✅ **6 data sources** integrated, validated and published
- ✅ **1500 wheat parcels** in CVL over 5 years
- ✅ **~620 MB** of structured data, publicly accessible
- ✅ **19 PDF documents** documenting the methodology
- ✅ **Critical bugs** identified and fixed (ET0, GEE empty windows)
- ✅ **Major optimizations** (SoilGrids WCS 91× faster)
- ✅ **Agronomic validation** on all sources (ranges, signatures)

21.2 What remains to be done

- 🔲 Master dataset construction (derived features + fusion) — 4-5 days
- 🔲 ML model training (5 phases) — 3-4 weeks
- 🔲 Interpretability analysis (SHAP, attention) — 1-2 weeks
- 🔲 Visualization dashboard (Streamlit) — 1 week
- 🔲 Final thesis writing — 4-6 weeks
- 🔲 Defense

21.3 Longer-term perspectives

Beyond the thesis:

- **"Data paper" publication** in Scientific Data (Nature) or ESSD (Copernicus) — the original character (first public multi-source dataset at parcel scale in France) justifies a publication
- **Extension to other regions** (Picardy, Champagne, Aquitaine) → test generalizability
- **Extension to other crops** (rapeseed, barley, corn)
- **Collaboration with cooperatives** to obtain real parcel-level harvest dates
- **Operational dashboard** for farmers or cooperatives

21.4 Final word

This thesis project represents several months of methodical work on complex and heterogeneous data. Beyond the final ML result, its main scientific contribution is **the public opening of a unique multi-source dataset**, accompanied by exhaustive documentation enabling its reuse by the community.

The accumulated experience — bugs, optimizations, decisions, trade-offs — is itself a precious asset, capitalized in this report to serve future students and researchers working on similar agricultural remote sensing projects.

Recommended citation:
Kassi R. (2025). *Master Dataset Report: Multi-source Sentinel-1/2 + climate + soil dataset for winter wheat harvest prediction in Centre-Val de Loire, France (2020-2024)*. Master's thesis methodology report, Indrashil University. Supervisors: Dr. V. Thakar, Dr. D. Ammar. Available on Kaggle: `kaggle.com/rgislikassi`

Master Dataset Report v1.0 — Master's thesis on winter wheat harvest window prediction
Centre-Val de Loire · Indrashil University · 2025 · Supervisors: Dr. Vishvjit Thakar, Dr. Doreid Ammar